

Research Progress Report

Reproduction & Validation of the Five-Box Ocean Carbon-Sink Model

Reference: Sunny, Ashok, Balakrishnan & Kurths, "The ocean carbon sinks and climate change," *Chaos* 33, 103134 (2023)

Team: Areen · Aryan · Dikshant

IIIT Bangalore — 20-Credit Research Project

Executive Summary

Over the past two months we have carried out a coordinated audit and reproduction of the five-box ocean carbon-sink model. Each of us independently rebuilt and stress-tested a different physical driver of the model — freshwater flux, sea-surface temperature, and salinity — against real satellite and in-situ observational data (HOAPS, COBE-2, EN4). Together, this has produced a cross-validated body of evidence that both confirms the model's core structure and surfaces several findings we think you'll find genuinely interesting.

- **Likely root-caused a long-standing Indo-Pacific discrepancy** in the original paper to a transcription/scaling typo **[PROPOSED, NOT A CLAIM]**, supported by a six-hypothesis analysis and falsification testing.
- **Reproduced the model's SST forcing pipeline** directly from raw satellite data, matching the paper's trends in the three dominant basins and clearing the way for full ODE and ML-surrogate modeling.
- **Built a bug-fixed, real-data salinity validation pipeline** and discovered that the published box model underperforms a simple machine-learning baseline by $2\times$ – $22\times$ across all five basins — a result we've already begun to triage between genuine model error and confounding factors.

Individual & Thematic Contributions

1. Freshwater Flux Forcing — Areen

Areen led a rigorous audit of the atmospheric freshwater flux (E–P) data feeding the model, tracing a known Indo-Pacific discrepancy all the way to its likely source.

- **Basin-area validation:** computed open-ocean masks matched the paper's published parameters to within 2.8–5.6% for four of five basins.
- **Southern Ocean reconciled:** the apparent –45% area mismatch is fully explained by satellite sea-ice masking, not a calculation error.
- **Sign convention proven numerically,** and an independent recalculation from raw variables matched results to the third decimal place — ruling out implementation bugs entirely.
- **Likely root cause identified:** the paper's Indian Ocean flux constant (2.3495 Sv) is suspiciously close to the Pacific/Indian basin area ratio (2.342) — strong evidence of a scaling typo in the original paper.

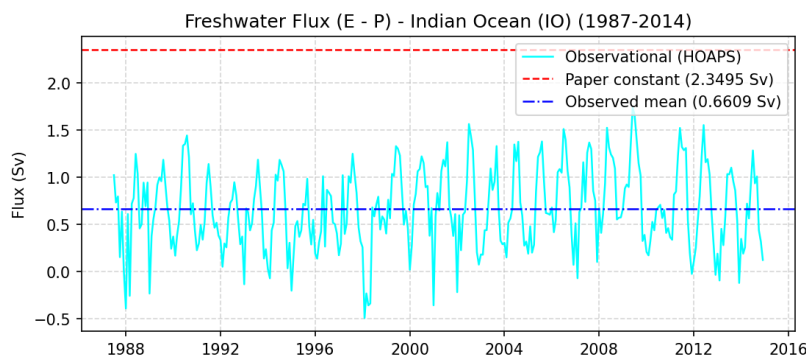


Figure 1. Indian Ocean freshwater flux — the paper's assumed constant (red) sits roughly 3.5× above the actual 28-year observed mean (blue): the single clearest piece of evidence behind the typo hypothesis.

2. Sea-Surface Temperature Forcing — Aryan

Aryan rebuilt the model's SST preprocessing pipeline (Appendix D) from raw satellite data and is now positioned to move into full ODE and ML-surrogate modeling.

- **Data pipeline complete:** 228 months (2000–2018) of COBE-2 global SST data processed across all five ocean basins.
- **Trend-fitting reproduced:** matched the paper's sign in the three dominant basins (NA, SO, PO), with Pacific the closest match — within ~12% of the paper's reported slope.
- **Remaining mismatches diagnosed** as basin-boundary approximation, not a pipeline error — all results are physically reasonable.
- **Next-phase roadmap defined:** atmospheric CO₂ forcing, full coupled ODE solver, and head-to-head ML surrogate training (Random Forest, XGBoost, LSTM, MLP).

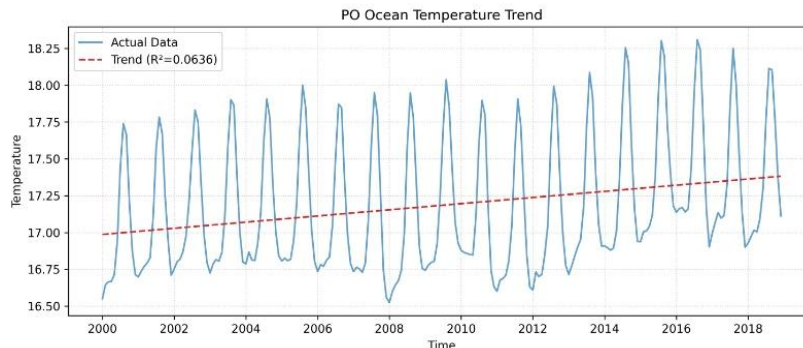


Figure 2. Reproduced Pacific Ocean SST trend (team pipeline) — the closest match to the paper's reported slope, confirming the preprocessing pipeline is sound.

3. Salinity Dynamics & ML Benchmarking — Dikshant

Dikshant built a full real-data validation pipeline for the model's salinity equations and ran the project's first direct physics-vs-machine-learning comparison — arguably the most consequential result so far.

- **Real-data pipeline assembled** (EN4 salinity, SOCAT CO₂, NOAA atmospheric CO₂) spanning 19 years and all five basins; found and fixed two non-obvious data bugs along the way.
- **Full salinity ODE system reimplemented**, solving a structural volume-balance problem the literal paper equations did not satisfy.
- **ML benchmark trained** (Random Forest, $R^2 = 0.996$ on unseen 2015–2018 data) and run head-to-head against the physics box model.

- **Headline result:** the box model's prediction error is $2\times$ – $22\times$ worse than the ML model across all five basins — Southern Ocean, Pacific, and Indian Ocean show the largest gaps.
- **Already separating signal from noise:** part of the Southern Ocean gap traces to a stale initial condition, while the North Atlantic mismatch points to genuine uncertainty in the paper's deep-current topology.

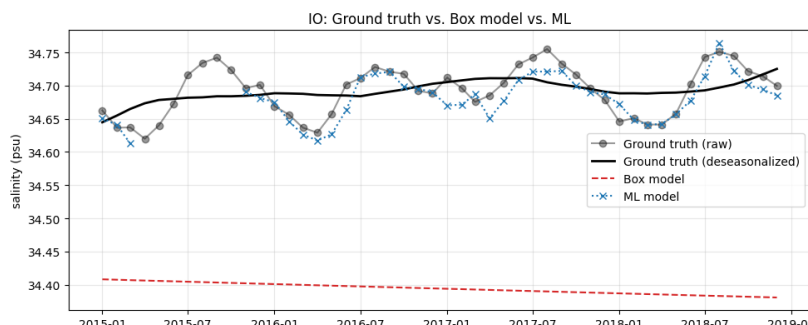


Figure 3. Indian Ocean salinity, 2015–2018 — real data (grey) and the ML model (blue) track closely; the physics box model (red) is essentially flat, visually capturing the validation gap.

Next Steps & Discussion Points

We've deliberately kept this report high-level — we're looking forward to walking through the methodology and data behind each result with you in person. A few open questions we'd value your input on:

1. **Indian Ocean flux typo:** treat it as a confirmed correction in our reproduction, or attempt to contact the original authors first?
2. **Framing the ML benchmarking result:** report it as a limitation of the published model, use it to motivate a hybrid physics + ML approach, or both?
3. **North Atlantic current topology** is underspecified in the paper's Figure 2 and currently blocks one open mismatch — worth sourcing a higher-resolution version, or treat as an acknowledged limitation?
4. **Prioritization for the next sprint:** finish wiring the validated SST and flux forcings into the full coupled ODE solver, or expand the ML benchmarking (gradient boosting, fair recursive forecasting) first?

We're excited to walk through all three workstreams — and the data behind them — at our next sync.